

МІНІСТЕРСТВО ОСВІТИ І НАУКИ УКРАЇНИ

Харківський національний університет імені В.Н. Каразіна

Навчально-науковий інститут комп'ютерних наук та штучного інтелекту

Кафедра математичного моделювання та аналізу даних

ПРАКТИЧНА РОБОТА № 2

на тему: «Використання нормалізації і вдосконалення програми для обробки великого обсягу даних»

Виконав: студент 3 курсу групи кс31
Спеціальності 124 «Комп'ютерні науки»
Кірієнко Данило

Прийняв: к.т.н., доцент каф. ММ та АД
Коробчинський К.П.

Мета роботи:

Доповнити програму під нові умови

Постановка задачі:

У другій частині завдання ви повинні змінити свою програму, щоб адаптуватись до змін у форматі вхідних даних. Очікується, що ви будете дотримуватись строгого формату виведення. Нагадаємо, що вихідний файл `students.txt` мав такий формат:

`StLastName, StFirstName, Grade, Classroom, Bus, TLastName, TFirstName`

Взагалі кажучи, якщо кожен учитель викладає саме в одному класі це займає багато місця, щоб поставити ім'я вчителя проти імені кожного учня. Тому тепер ми переглядаємо формат вхідного файлу. Тепер оригінальний `students.txt` замінюється парою файлів: `list.txt` та `teacher.txt`.

Файл `list.txt` має такий формат:

`StLastName, StFirstName, Grade, Classroom, Bus`

Зразок рядка файла `list.txt`:

`DROP, SHERMAN, 0, 104, 53`

Це можна інтерпретувати наступним чином: Sherman DROP учень, який навчається у класі (номер класної кімнати) 104, та їде автобусом № 53. Формат `teacher.txt` має вигляд:

TLastName, TFirstName, Classroom

Рядок з файла teachers.txt має вигляд:

NIBLER, JERLENE, 104

Це можна інтерпретувати наступним чином: NIBLER Jerlene викладає у класі (номер класної кімнати) 104.

Файли даних доступні сайті університету, прямі URL-адреси для файлів:
<https://stm.khai.edu/list.txt> та teachers.txt.

Інформація в цих файлах точно така ж, як і у вихідному файлі students.txt, тому відповіді на тестові пошуки в обох версіях вашої програми повинні бути однаковими.

Ваша мета – змінити програму шкільного пошуку з першої практичної роботи, для обробки нових форматів вхідних даних. Крім того, вам пропонується розробити та впровадити у новій програмі додаткові засоби пошуку.

Під час виконання необхідно скопіювати результати роботи першої практичної роботи та порівняти з результатами другої. Також обов'язково необхідно додати час виконання кожного з

запитів.

Результат виконання:

Для виконання консольної роботи я використовував мову C++.

```

E:\UknordShe\BH3\obd\3-obd\2\cmake-build-debug\2.exe
Enter command:$ COOKUS
COOKUS, XUAN - Grade: 3, Classroom: 107, Bus: 52, Teachers: FAFARD ROCIO; BODZIONY LUZ; HANTZ GAVIN;
STEIB ADOLPH; ALPERT ADOLPH; GAMBREL NANCY; COOL JONATHAN; KERBS JED; NISTENDIRK NANCY; KERBS PERLA; STEI
BODZIONY NANCY; COOL GAVIN; FAFARD LUZ; BODZIONY JAE; GAMBREL JONATHAN; NISTENDIRK GAVIN; BODZIONY PERLA;
REUBEN; NISTENDIRK BENITO; GAMBREL ROCIO; ALPERT NANCY; STEIB JED; ALPERT BENITO; FAFARD REUBEN; COOL RE
IONCHIO JONATHAN; CHIONCHIO BENITO; HAMER GAVIN; ALPERT GAVIN; COOL PERLA; ALPERT PERLA; HANTZ JAE; CHION
CIO; CHIONCHIO JED; CHIONCHIO GAVIN; FALKER ROCIO; FALKER BENITO;
COOKUS, XUAN - Grade: 3, Classroom: 104, Bus: 55, Teachers: STEIB GALE; KERBS JONATHAN; STEIB JAE; BODZIONY
DZIONY ROCIO; KERBS JED; COOL REUBEN; FALKER GAVIN; COOL LUZ; HAMER ROCIO; HAMER GAVIN; ALPERT LUZ; HAME
N; CHIONCHIO JAE; ALPERT REUBEN; NISTENDIRK JONATHAN; GAMBREL REUBEN; CHIONCHIO JED; ALPERT PERLA; HAMER
BREL JONATHAN; NISTENDIRK JED; ALPERT ROCIO; ALPERT NANCY; HAMER JONATHAN; COOL ADOLPH; BODZIONY PERLA; H
NANCY; HANTZ JONATHAN; CHIONCHIO REUBEN; KERBS GAVIN; STEIB PERLA; HANTZ ROCIO; NISTENDIRK GAVIN; ALPERT JO
NISTENDIRK ADOLPH; FALKER JAE; HANTZ BENITO; HANTZ JAE; FAFARD JAE; FAFARD PERLA; HAMER PERLA; FAFARD GA
BS REUBEN; BODZIONY JONATHAN; HANTZ JED; HANTZ NANCY;
COOKUS, MANIE - Grade: 5, Classroom: 112, Bus: 53, Teachers: CHIONCHIO PERLA; FAFARD PERLA; FALKER ADOLPH; G
JONATHAN; FALKER PERLA; NISTENDIRK ADOLPH; COOL ROCIO; BODZIONY BENITO; COOL NANCY; HAMER NANCY; FALKER J
BODZIONY LUZ; KERBS JED; HANTZ ROCIO; STEIB PERLA; COOL REUBEN; CHIONCHIO BENITO; ALPERT JONATHAN; FAFAR
; CHIONCHIO GAVIN; STEIB JONATHAN; HANTZ ADOLPH; COOL PERLA; KERBS JAE; ALPERT GAVIN; COOL JAE; CHIONCHIO
LPERT JED; HANTZ NANCY; HAMER BENITO; FALKER NANCY; STEIB BENITO; ALPERT PERLA; BODZIONY ADOLPH; KERBS G
EIB JAE; BODZIONY NANCY; HAMER JAE; STEIB LUZ; HANTZ REUBEN; ALPERT LUZ; GAMBREL PERLA;
COOKUS, WAN - Grade: 5, Classroom: 109, Bus: 56, Teachers: GAMBREL JAE; CHIONCHIO JONATHAN; CHIONCHIO GAVIN;
JED; KERBS JED; FAFARD GAVIN; CHIONCHIO BENITO; NISTENDIRK JED; FALKER GAVIN; BODZIONY ADOLPH; FAFARD ROC
RT PERLA; GAMBREL JONATHAN; HANTZ BENITO; ALPERT JED; HAMER ADOLPH; FAFARD PERLA; HAMER PERLA; COOL PERL
JONATHAN; HANTZ REUBEN; FAFARD JAE; FALKER NANCY; BODZIONY JED; HANTZ LUZ; NISTENDIRK ADOLPH; CHIONCHIO
AMER GAVIN; HAMER BENITO; STEIB GAVIN; HANTZ PERLA; BODZIONY JONATHAN; NISTENDIRK JONATHAN; KERBS LUZ; GA
OCIO; HANTZ ROCIO; FALKER JONATHAN; CHIONCHIO JAE; STEIB JONATHAN; ALPERT NANCY; GAMBREL ADOLPH; STEIB JA
NANCY; HANTZ JONATHAN; GAMBREL BENITO;
COOKUS, RANDOLPH - Grade: 5, Classroom: 111, Bus: 52, Teachers: NISTENDIRK NANCY; GAMBREL PERLA; NISTENDIRK
ANTZ JED; CHIONCHIO BENITO; HANTZ PERLA; KERBS JAE; ALPERT LUZ; ALPERT PERLA; NISTENDIRK ADOLPH; ALPERT
; STEIB JAE; FALKER NANCY; FALKER JED; KERBS JONATHAN; FAFARD JAE; COOL PERLA; HANTZ ADOLPH; HAMER JAE;
GAVIN; STEIB JONATHAN; GAMBREL NANCY; HAMER PERLA; NISTENDIRK JAE; HANTZ REUBEN; ALPERT JED; KERBS BENITO
BENITO; STEIB NANCY; COOL GAVIN; GAMBREL JONATHAN; NISTENDIRK JED; STEIB ADOLPH; STEIB LUZ; BODZIONY PE
L ROCIO; HAMER JED; CHIONCHIO LUZ; HAMER NANCY; KERBS NANCY; HANTZ JAE; NISTENDIRK LUZ;
COOKUS, JANNETTE - Grade: 5, Classroom: 105, Bus: 51, Teachers: HANTZ JED; BODZIONY PERLA; FAFARD JED; ALPER
O; FAFARD ROCIO; COOL JED; COOL JONATHAN; FAFARD GAVIN; KERBS ADOLPH; COOL ADOLPH; STEIB JED; NISTENDIRK
FAFARD JAE; KERBS NANCY; STEIB BENITO; KERBS PERLA; HANTZ ROCIO; COOL JAE; BODZIONY JAE; ALPERT ADOLPH;
IO GAVIN; NISTENDIRK REUBEN; GAMBREL JAE; NISTENDIRK ROCIO; KERBS LUZ; FAFARD BENITO; BODZIONY NANCY; FAL
; NISTENDIRK JAE; ALPERT NANCY; HANTZ JAE; HAMER JED; FALKER GAVIN; ALPERT PERLA; NISTENDIRK JED; STEIB
ANTZ REUBEN; COOL GAVIN; KERBS JAE; FALKER ROCIO; STEIB REUBEN; BODZIONY JED; FALKER BENITO; FAFARD NANC
T GAVIN; BODZIONY LUZ; BODZIONY BENITO; FALKER PERLA; CHIONCHIO BENITO; HAMER ADOLPH; FALKER NANCY; NISTE
ENITO; NISTENDIRK JONATHAN; STEIB GAVIN; COOL PERLA; CHIONCHIO REUBEN; FAFARD JONATHAN;
COOKUS, INGER - Grade: 2, Classroom: 110, Bus: 54, Teachers: ALPERT JONATHAN; FALKER REUBEN; CHIONCHIO LUZ;
ERLA; KERBS JAE; HAMER JONATHAN; ALPERT JAE; HAMER GAVIN; GAMBREL BENITO; GAMBREL REUBEN; ALPERT ADOLPH;
PERLA; NISTENDIRK JED; FALKER JED; STEIB REUBEN; BODZIONY JED; FAFARD BENITO; FAFARD ADOLPH; GAMBREL JAE
BENITO; COOL PERLA; NISTENDIRK ROCIO; HANTZ JAE; COOL ROCIO; HANTZ REUBEN; STEIB NANCY; ALPERT LUZ; COOL
LPERT PERLA; HAMER PERLA; ALPERT JED; COOL JAE; CHIONCHIO ADOLPH; BODZIONY JAE; BODZIONY LUZ; NISTENDIRK
HAMER ADOLPH; ALPERT ROCIO; HAMER REUBEN; CHIONCHIO JONATHAN; FALKER GAVIN; NISTENDIRK PERLA; HAMER BENI
REL NANCY; FAFARD PERLA; KERBS BENITO;
COOKUS, TOMAS - Grade: 4, Classroom: 109, Bus: 54, Teachers: GAMBREL JAE; CHIONCHIO JONATHAN; CHIONCHIO GAVI
JED; KERBS JED; FAFARD GAVIN; CHIONCHIO BENITO; NISTENDIRK JED; FALKER GAVIN; BODZIONY ADOLPH; FAFARD PE
PERT PERLA; GAMBREL JONATHAN; HANTZ BENITO; ALPERT JED; HAMER ADOLPH; FAFARD PERLA; HAMER PERLA; COOL PER
L JONATHAN; HANTZ REUBEN; FAFARD JAE; FALKER NANCY; BODZIONY JED; HANTZ LUZ; NISTENDIRK ADOLPH; CHIONCHIO
HAMER GAVIN; HAMER BENITO; STEIB GAVIN; HANTZ PERLA; BODZIONY JONATHAN; NISTENDIRK JONATHAN; KERBS LUZ;
ROCIO; HANTZ ROCIO; FALKER JONATHAN; CHIONCHIO JAE; STEIB JONATHAN; ALPERT NANCY; GAMBREL ADOLPH; STEIB
L NANCY; HANTZ JONATHAN; GAMBREL BENITO;
COOKUS, ZANDRA - Grade: 6, Classroom: 106, Bus: 51, Teachers: BODZIONY LUZ; GAMBREL JONATHAN; HAMER ROCIO; H
Z; NISTENDIRK ADOLPH; FAFARD ADOLPH; STEIB NANCY; HAMER GAVIN; HANTZ GAVIN; COOL JAE; COOL NANCY; BODZION
; KERBS JED; HANTZ NANCY; FALKER BENITO; FAFARD REUBEN; ALPERT ROCIO; HANTZ BENITO; GAMBREL ROCIO; CHIONC
ATHAN; COOL JONATHAN; BODZIONY BENITO; FALKER PERLA; ALPERT JED; FALKER JAE; COOL ROCIO; ALPERT GAVIN; FA
OLPH; FAFARD PERLA; GAMBREL ADOLPH; HANTZ JAE; HANTZ ADOLPH; CHIONCHIO JED; ALPERT JONATHAN; CHIONCHIO GA
BREL PERLA; COOL PERLA; FALKER NANCY; KERBS GAVIN; FALKER JED; CHIONCHIO JAE; HANTZ LUZ; NISTENDIRK BENI
B JONATHAN; COOL BENITO; STEIB JED;
COOKUS, HYE - Grade: 5, Classroom: 107, Bus: 56, Teachers: FAFARD ROCIO; BODZIONY LUZ; HANTZ GAVIN; STEIB A
LPERT ADOLPH; GAMBREL NANCY; COOL JONATHAN; KERBS JED; NISTENDIRK NANCY; KERBS PERLA; STEIB LUZ; BODZIONY
COOL GAVIN; FAFARD LUZ; BODZIONY JAE; GAMBREL JONATHAN; NISTENDIRK GAVIN; BODZIONY PERLA; FALKER REUBEN;
IRK BENITO; GAMBREL ROCIO; ALPERT NANCY; STEIB JED; ALPERT BENITO; FAFARD REUBEN; COOL REUBEN; CHIONCHIO
N; CHIONCHIO BENITO; HAMER GAVIN; ALPERT GAVIN; COOL PERLA; ALPERT PERLA; HANTZ JAE; CHIONCHIO ROCIO; CHI
JED; CHIONCHIO GAVIN; FALKER ROCIO; FALKER BENITO;
COOKUS, XUAN - Grade: 1, Classroom: 105, Bus: 52, Teachers: HANTZ JED; BODZIONY PERLA; FAFARD JED; ALPERT B
AFARD ROCIO; COOL JED; COOL JONATHAN; FAFARD GAVIN; KERBS ADOLPH; COOL ADOLPH; STEIB JED; NISTENDIRK ADOL
ARD JAE; KERBS NANCY; STEIB BENITO; KERBS PERLA; HANTZ ROCIO; COOL JAE; BODZIONY JAE; ALPERT ADOLPH; CHI
GAVIN; NISTENDIRK REUBEN; GAMBREL JAE; NISTENDIRK ROCIO; KERBS LUZ; FAFARD BENITO; BODZIONY NANCY; FALKER
STENDIRK JAE; ALPERT NANCY; HANTZ JAE; HAMER JED; FALKER GAVIN; ALPERT PERLA; NISTENDIRK JED; STEIB ROCI
REUBEN; COOL GAVIN; KERBS JAE; FALKER ROCIO; STEIB REUBEN; BODZIONY JED; FALKER BENITO; FAFARD NANCY; A
AVIN; BODZIONY LUZ; BODZIONY BENITO; FALKER PERLA; CHIONCHIO BENITO; HAMER ADOLPH; FALKER NANCY; NISTENDIR
O; NISTENDIRK JONATHAN; STEIB GAVIN; COOL PERLA; CHIONCHIO REUBEN; FAFARD JONATHAN;
COOKUS, INGER - Grade: 5, Classroom: 103, Bus: 51, Teachers: FALKER ADOLPH; CHIONCHIO GAVIN; FAFARD ROCIO; H
D; COOL REUBEN; NISTENDIRK JED; KERBS PERLA; NISTENDIRK GAVIN; BODZIONY LUZ; ALPERT JONATHAN; HAMER REUBE

```

teachers.txt

129:16 CRLF UTF-8 4

Рисунок 1 - Скріншот виконання пошуку по прізвищу

```

Z; NISTENDIRK ADOLPH; FAFARD ADOLPH; STEIB NANCY; HAMER GAVIN; HANTZ GAVIN; COOL JAE; COOL NANCY; BODZIONY RO
; KERBS JED; HANTZ NANCY; FALKER BENITO; FAFARD REUBEN; ALPERT ROCIO; HANTZ BENITO; GAMBREL ROCIO; CHIONCHIO
ATHAN; COOL JONATHAN; BODZIONY BENITO; FALKER PERLA; ALPERT JED; FALKER JAE; COOL ROCIO; ALPERT GAVIN; FALKER
OLPH; FAFARD PERLA; GAMBREL ADOLPH; HANTZ JAE; HANTZ ADOLPH; CHIONCHIO JED; ALPERT JONATHAN; CHIONCHIO GAVIN;
BREL PERLA; COOL PERLA; FALKER NANCY; KERBS GAVIN; FALKER JED; CHIONCHIO JAE; HANTZ LUZ; NISTENDIRK BENITO; S
B JONATHAN; COOL BENITO; STEIB JED;
COOKUS, HYE - Grade: 5, Classroom: 107, Bus: 56, Teachers: FAFARD ROCIO; BODZIONY LUZ; HANTZ GAVIN; STEIB ADOLPH
LPERT ADOLPH; GAMBREL NANCY; COOL JONATHAN; KERBS JED; NISTENDIRK NANCY; KERBS PERLA; STEIB LUZ; BODZIONY NANC
COOL GAVIN; FAFARD LUZ; BODZIONY JAE; GAMBREL JONATHAN; NISTENDIRK GAVIN; BODZIONY PERLA; FALKER REUBEN; NIST
IRK BENITO; GAMBREL ROCIO; ALPERT NANCY; STEIB JED; ALPERT BENITO; FAFARD REUBEN; COOL REUBEN; CHIONCHIO JONAT
N; CHIONCHIO BENITO; HAMER GAVIN; ALPERT GAVIN; COOL PERLA; ALPERT PERLA; HANTZ JAE; CHIONCHIO ROCIO; CHIONCHIO
JED; CHIONCHIO GAVIN; FALKER ROCIO; FALKER BENITO;
COOKUS, XUAN - Grade: 1, Classroom: 105, Bus: 52, Teachers: HANTZ JED; BODZIONY PERLA; FAFARD JED; ALPERT BENITO
AFARD ROCIO; COOL JED; COOL JONATHAN; FAFARD GAVIN; KERBS ADOLPH; COOL ADOLPH; STEIB JED; NISTENDIRK ADOLPH;
ARD JAE; KERBS NANCY; STEIB BENITO; KERBS PERLA; HANTZ ROCIO; COOL JAE; BODZIONY JAE; ALPERT ADOLPH; CHIONCHIO
GAVIN; NISTENDIRK REUBEN; GAMBREL JAE; NISTENDIRK ROCIO; KERBS LUZ; FAFARD BENITO; BODZIONY NANCY; FALKER JED;
STENDIRK JAE; ALPERT NANCY; HANTZ JAE; HAMER JED; FALKER GAVIN; ALPERT PERLA; NISTENDIRK JED; STEIB ROCIO; HAM
REUBEN; COOL GAVIN; KERBS JAE; FALKER ROCIO; STEIB REUBEN; BODZIONY JED; FALKER BENITO; FAFARD NANCY; ALPERT
AVIN; BODZIONY LUZ; BODZIONY BENITO; FALKER PERLA; CHIONCHIO BENITO; HAMER ADOLPH; FALKER NANCY; NISTENDIRK BEN
O; NISTENDIRK JONATHAN; STEIB GAVIN; COOL PERLA; CHIONCHIO REUBEN; FAFARD JONATHAN;
COOKUS, INGER - Grade: 5, Classroom: 103, Bus: 51, Teachers: FALKER ADOLPH; CHIONCHIO GAVIN; FAFARD ROCIO; HANTZ
D; COOL REUBEN; NISTENDIRK JED; KERBS PERLA; NISTENDIRK GAVIN; BODZIONY LUZ; ALPERT JONATHAN; HAMER REUBEN; AL
T REUBEN; ALPERT LUZ; COOL GAVIN; CHIONCHIO PERLA; STEIB LUZ; KERBS ADOLPH; STEIB GAVIN; STEIB JAE; NISTENDIRK
ADOLPH; HANTZ JAE; COOL BENITO; FALKER JED; KERBS REUBEN; COOL ROCIO; CHIONCHIO NANCY; FAFARD NANCY; NISTENDIRK
LUZ; FAFARD JONATHAN; GAMBREL LUZ; GAMBREL PERLA; HANTZ BENITO; HANTZ ADOLPH; HAMER ROCIO; KERBS JONATHAN; COOL
JAE; FALKER GAVIN; COOL LUZ; NISTENDIRK JAE; BODZIONY JAE; COOL PERLA; STEIB ROCIO; HANTZ REUBEN; KERBS BENITO
BODZIONY JONATHAN; CHIONCHIO JAE; FAFARD REUBEN;
COOKUS, DICK - Grade: 2, Classroom: 104, Bus: 54, Teachers: STEIB GALE; KERBS JONATHAN; STEIB JAE; BODZIONY JED;
DZIONY ROCIO; KERBS JED; COOL REUBEN; FALKER GAVIN; COOL LUZ; HAMER ROCIO; HAMER GAVIN; ALPERT LUZ; HAMER RE
N; CHIONCHIO JAE; ALPERT REUBEN; NISTENDIRK JONATHAN; GAMBREL REUBEN; CHIONCHIO JED; ALPERT PERLA; HAMER LUZ;
BREL JONATHAN; NISTENDIRK JED; ALPERT ROCIO; ALPERT NANCY; HAMER JONATHAN; COOL ADOLPH; BODZIONY PERLA; HAMER
NCY; HANTZ JONATHAN; CHIONCHIO REUBEN; KERBS GAVIN; STEIB PERLA; HANTZ ROCIO; NISTENDIRK GAVIN; ALPERT JONATHAN
NISTENDIRK ADOLPH; FALKER JAE; HANTZ BENITO; HANTZ JAE; FAFARD JAE; FAFARD PERLA; HAMER PERLA; FAFARD GAVIN;
BS REUBEN; BODZIONY JONATHAN; HANTZ JED; HANTZ NANCY;
COOKUS, MANIE - Grade: 2, Classroom: 102, Bus: 53, Teachers: KERBS BENITO; HANTZ JED; CHIONCHIO PERLA; CHIONCHIO
CIO; COOL ROCIO; FAFARD REUBEN; KERBS JAE; ALPERT NANCY; CHIONCHIO ADOLPH; ALPERT GAVIN; FALKER GAVIN; KERBS
IN; GAMBREL NANCY; HAMER GAVIN; KERBS JONATHAN; STEIB ADOLPH; BODZIONY LUZ; ALPERT PERLA; STEIB NANCY; HAMER
CY; HAMER ROCIO; CHIONCHIO LUZ; GAMBREL PERLA; KERBS NANCY; BODZIONY JAE; COOL PERLA; CHIONCHIO JED; HANTZ NANCY
; FALKER ROCIO; HANTZ GAVIN; FAFARD LUZ; KERBS PERLA; CHIONCHIO JAE; GAMBREL JONATHAN; NISTENDIRK REUBEN; FALKER
REUBEN; NISTENDIRK JED; NISTENDIRK JAE; HANTZ REUBEN; ALPERT JED; KERBS ADOLPH; STEIB GAVIN; NISTENDIRK ADOLPH
HANTZ PERLA; COOL JONATHAN; FALKER JED; FALKER NANCY; GAMBREL LUZ; HAMER PERLA;
COOKUS, TAMESHA - Grade: 3, Classroom: 104, Bus: 53, Teachers: STEIB GALE; KERBS JONATHAN; STEIB JAE; BODZIONY J
BODZIONY ROCIO; KERBS JED; COOL REUBEN; FALKER GAVIN; COOL LUZ; HAMER ROCIO; HAMER GAVIN; ALPERT LUZ; HAMER
UBEN; CHIONCHIO JAE; ALPERT REUBEN; NISTENDIRK JONATHAN; GAMBREL REUBEN; CHIONCHIO JED; ALPERT PERLA; HAMER LUZ
GAMBREL JONATHAN; NISTENDIRK JED; ALPERT ROCIO; ALPERT NANCY; HAMER JONATHAN; COOL ADOLPH; BODZIONY PERLA; HAMER
NANCY; HANTZ JONATHAN; CHIONCHIO REUBEN; KERBS GAVIN; STEIB PERLA; HANTZ ROCIO; NISTENDIRK GAVIN; ALPERT JONAT
N; NISTENDIRK ADOLPH; FALKER JAE; HANTZ BENITO; HANTZ JAE; FAFARD JAE; FAFARD PERLA; HAMER PERLA; FAFARD GAVIN
KERBS REUBEN; BODZIONY JONATHAN; HANTZ JED; HANTZ NANCY;
COOKUS, BILLY - Grade: 3, Classroom: 111, Bus: 51, Teachers: NISTENDIRK NANCY; GAMBREL PERLA; NISTENDIRK GAVIN; H
Z JED; CHIONCHIO BENITO; HANTZ PERLA; KERBS JAE; ALPERT LUZ; ALPERT PERLA; NISTENDIRK ADOLPH; ALPERT JONATHAN
TEIB JAE; FALKER NANCY; FALKER JED; KERBS JONATHAN; FAFARD JAE; COOL PERLA; HANTZ ADOLPH; HAMER JAE; FALKER
IN; STEIB JONATHAN; GAMBREL NANCY; HAMER PERLA; NISTENDIRK JAE; HANTZ REUBEN; ALPERT JED; KERBS BENITO; FAFARD
ENITO; STEIB NANCY; COOL GAVIN; GAMBREL JONATHAN; NISTENDIRK JED; STEIB ADOLPH; STEIB LUZ; BODZIONY PERLA; COOL
ROCIO; HAMER JED; CHIONCHIO LUZ; HAMER NANCY; KERBS NANCY; HANTZ JAE; NISTENDIRK LUZ;
COOKUS, BILLY - Grade: 3, Classroom: 111, Bus: 55, Teachers: NISTENDIRK NANCY; GAMBREL PERLA; NISTENDIRK GAVIN; H
Z JED; CHIONCHIO BENITO; HANTZ PERLA; KERBS JAE; ALPERT LUZ; ALPERT PERLA; NISTENDIRK ADOLPH; ALPERT JONATHAN
TEIB JAE; FALKER NANCY; FALKER JED; KERBS JONATHAN; FAFARD JAE; COOL PERLA; HANTZ ADOLPH; HAMER JAE; FALKER
IN; STEIB JONATHAN; GAMBREL NANCY; HAMER PERLA; NISTENDIRK JAE; HANTZ REUBEN; ALPERT JED; KERBS BENITO; FAFARD
ENITO; STEIB NANCY; COOL GAVIN; GAMBREL JONATHAN; NISTENDIRK JED; STEIB ADOLPH; STEIB LUZ; BODZIONY PERLA; COOL
ROCIO; HAMER JED; CHIONCHIO LUZ; HAMER NANCY; KERBS NANCY; HANTZ JAE; NISTENDIRK LUZ;
COOKUS, BILLY - Grade: 6, Classroom: 109, Bus: 51, Teachers: GAMBREL JAE; CHIONCHIO JONATHAN; CHIONCHIO GAVIN; HAM
JED; KERBS JED; FAFARD GAVIN; CHIONCHIO BENITO; NISTENDIRK JED; FALKER GAVIN; BODZIONY ADOLPH; FAFARD ROCIO;
PERT PERLA; GAMBREL JONATHAN; HANTZ BENITO; ALPERT JED; HAMER ADOLPH; FAFARD PERLA; HAMER PERLA; COOL PERLA;
L JONATHAN; HANTZ REUBEN; FAFARD JAE; FALKER NANCY; BODZIONY JED; HANTZ LUZ; NISTENDIRK ADOLPH; CHIONCHIO ROCIO
HAMER GAVIN; HAMER BENITO; STEIB GAVIN; HANTZ PERLA; BODZIONY JONATHAN; NISTENDIRK JONATHAN; KERBS LUZ; GAMBREL
ROCIO; HANTZ ROCIO; FALKER JONATHAN; CHIONCHIO JAE; STEIB JONATHAN; ALPERT NANCY; GAMBREL ADOLPH; STEIB JAE;
L NANCY; HANTZ JONATHAN; GAMBREL BENITO;
Search Time: 0.164908 seconds
Enter command: 52
Error! Invalid command!

```

Рисунок 2 - Скріншот виконання пошуку по прізвищу

```

Enter command: 52
Error! Invalid command!
Enter command: B 52
COOKUS, XUAN - Bus: 52
ELHADDAD, SHANTE - Bus: 52
IMMERMAN, DEVIN - Bus: 52
OBREIN, STAN - Bus: 52
CORONADO, DIMPLE - Bus: 52
SARAO, DIEDRA - Bus: 52
EARLY, TORY - Bus: 52
KEMERER, ROSETTA - Bus: 52
BREVO, DICK - Bus: 52
CREMEANS, RANDOLPH - Bus: 52
BOYTER, TOMAS - Bus: 52
BERBES, MANIE - Bus: 52
BERBES, DICK - Bus: 52
CORKER, RANDOLPH - Bus: 52
CORKER, TAMESHA - Bus: 52
BERBES, CARTER - Bus: 52
CREMEANS, FLOY - Bus: 52
COVINGTON, ZANDRA - Bus: 52
COOKUS, RANDOLPH - Bus: 52
CLECKLER, INGER - Bus: 52
COMO, MANIE - Bus: 52
BUSSMANN, JANNETTE - Bus: 52
BERBES, JANNETTE - Bus: 52
BEAN, WAN - Bus: 52
BERBES, WAN - Bus: 52
COMO, INGER - Bus: 52
BRODERSEN, ZANDRA - Bus: 52
CLECKLER, WAN - Bus: 52
BEAN, TOMAS - Bus: 52
CREMEANS, CARTER - Bus: 52
BEX, TAMESHA - Bus: 52
CORONADO, BILLY - Bus: 52
COMO, WAN - Bus: 52
BERBES, TOMAS - Bus: 52
BERBES, ZANDRA - Bus: 52
BUSSMANN, ZANDRA - Bus: 52
COOKUS, XUAN - Bus: 52
CIGANEK, ZANDRA - Bus: 52
BUSSMANN, TOMAS - Bus: 52
BEAN, JANNETTE - Bus: 52
CORKER, BILLY - Bus: 52
CORONADO, CARTER - Bus: 52
BREVO, BILLY - Bus: 52
CORKER, TAMESHA - Bus: 52
BERBES, XUAN - Bus: 52
BUSSMANN, TOMAS - Bus: 52
CREMEANS, WAN - Bus: 52
COMO, WAN - Bus: 52
BEAN, DICK - Bus: 52
COMO, ZANDRA - Bus: 52
BOYTER, MANIE - Bus: 52
CLECKLER, DICK - Bus: 52
CORKER, FLOY - Bus: 52
CLECKLER, TOMAS - Bus: 52
COVINGTON, MANIE - Bus: 52
BRODERSEN, JANNETTE - Bus: 52
CIGANEK, DICK - Bus: 52
BEAN, INGER - Bus: 52
BEX, DICK - Bus: 52
BEAN, WAN - Bus: 52
BERBES, INGER - Bus: 52
BUSSMANN, INGER - Bus: 52
COVINGTON, DIMPLE - Bus: 52
BEX, FLOY - Bus: 52
BOYTER, RANDOLPH - Bus: 52
BUSSMANN, XUAN - Bus: 52
BOYTER, TOMAS - Bus: 52
COVINGTON, MANIE - Bus: 52
BERBES, FLOY - Bus: 52
CORONADO, CARTER - Bus: 52
BEX, RANDOLPH - Bus: 52
BEAN, FLOY - Bus: 52
BEX, FLOY - Bus: 52
CORKER, BILLY - Bus: 52
COVINGTON, TAMESHA - Bus: 52
BEX, WAN - Bus: 52
CLECKLER, INGER - Bus: 52
BUSSMANN, CARTER - Bus: 52
CIGANEK, RANDOLPH - Bus: 52
BEX, INGER - Bus: 52
BUSSMANN, INGER - Bus: 52
CORKER, DIMPLE - Bus: 52
Search Time: 0.0344525 seconds
Enter command: C 107
COOKUS, XUAN - Classroom: 107
ELHADDAD, SHANTE - Classroom: 107

```

Рисунок 3 - Скріншоти виконання пошуку по автобусу

```
Enter command: C 107T
FAFARD ROCIO - Classroom: 107
BODZIONY LUZ - Classroom: 107
HANTZ GAVIN - Classroom: 107
STEIB ADOLPH - Classroom: 107
ALPERT ADOLPH - Classroom: 107
GAMBREL NANCY - Classroom: 107
COOL JONATHAN - Classroom: 107
KERBS JED - Classroom: 107
NISTENDIRK NANCY - Classroom: 107
KERBS PERLA - Classroom: 107
STEIB LUZ - Classroom: 107
BODZIONY NANCY - Classroom: 107
COOL GAVIN - Classroom: 107
FAFARD LUZ - Classroom: 107
BODZIONY JAE - Classroom: 107
GAMBREL JONATHAN - Classroom: 107
NISTENDIRK GAVIN - Classroom: 107
BODZIONY PERLA - Classroom: 107
FALKER REUBEN - Classroom: 107
NISTENDIRK BENITO - Classroom: 107
GAMBREL ROCIO - Classroom: 107
ALPERT NANCY - Classroom: 107
STEIB JED - Classroom: 107
ALPERT BENITO - Classroom: 107
FAFARD REUBEN - Classroom: 107
COOL REUBEN - Classroom: 107
CHIONCHIO JONATHAN - Classroom: 107
CHIONCHIO BENITO - Classroom: 107
HAMER GAVIN - Classroom: 107
ALPERT GAVIN - Classroom: 107
COOL PERLA - Classroom: 107
ALPERT PERLA - Classroom: 107
HANTZ JAE - Classroom: 107
CHIONCHIO ROCIO - Classroom: 107
CHIONCHIO JED - Classroom: 107
CHIONCHIO GAVIN - Classroom: 107
FALKER ROCIO - Classroom: 107
FALKER BENITO - Classroom: 107
Search Time: 0.0156971 seconds
Enter command: G 3
```

Рисунок 4 - Скріншот виконання пошуку вчителя по класній кімнаті


```

Enter command: C 107
COOKUS, XUAN - Classroom: 107
ELHADDAD, SHANTE - Classroom: 107
SWEDLUND, SHARRI - Classroom: 107
CIGANEK, MANIE - Classroom: 107
COVINGTON, TOMAS - Classroom: 107
EARLY, TORY - Classroom: 107
LINHART, LELA - Classroom: 107
BERBES, DICK - Classroom: 107
CORONADO, INGER - Classroom: 107
CORONADO, JANNETTE - Classroom: 107
CORKER, TAMESHA - Classroom: 107
BOYTER, WAN - Classroom: 107
BEX, JANNETTE - Classroom: 107
CIGANEK, JANNETTE - Classroom: 107
COMO, FLOY - Classroom: 107
CORKER, JANNETTE - Classroom: 107
COVINGTON, DICK - Classroom: 107
CLECKLER, BILLY - Classroom: 107
BUSSMANN, RANDOLPH - Classroom: 107
COOKUS, HYE - Classroom: 107
BEX, TAMESHA - Classroom: 107
BEAN, DICK - Classroom: 107
BEX, HYE - Classroom: 107
CORKER, ZANDRA - Classroom: 107
BUSSMANN, HYE - Classroom: 107
COMO, DIMPLE - Classroom: 107
COVINGTON, CARTER - Classroom: 107
BUSSMANN, XUAN - Classroom: 107
BEAN, FLOY - Classroom: 107
BREVO, RANDOLPH - Classroom: 107
BOYTER, XUAN - Classroom: 107
BOYTER, CARTER - Classroom: 107
BOYTER, RANDOLPH - Classroom: 107
CORKER, TAMESHA - Classroom: 107
Search Time: 0.0150397 seconds
Enter command: C 107T
FAFARD ROCIO - Classroom: 107
BODZIONY LUZ - Classroom: 107
HANTZ GAVIN - Classroom: 107
STEIB ADOLPH - Classroom: 107
ALPERT ADOLPH - Classroom: 107
GAMBREL NANCY - Classroom: 107
COOL JONATHAN - Classroom: 107
KERBS JED - Classroom: 107
NISTENDIRK NANCY - Classroom: 107
KERBS PERLA - Classroom: 107
STEIB LUZ - Classroom: 107
BODZIONY NANCY - Classroom: 107
COOL GAVIN - Classroom: 107
FAFARD LUZ - Classroom: 107
BODZIONY JAE - Classroom: 107
GAMBREL JONATHAN - Classroom: 107
NISTENDIRK GAVIN - Classroom: 107
BODZIONY PERLA - Classroom: 107
FALKER REUBEN - Classroom: 107
NISTENDIRK BENITO - Classroom: 107
GAMBREL ROCIO - Classroom: 107
ALPERT NANCY - Classroom: 107
STEIB JED - Classroom: 107
ALPERT BENITO - Classroom: 107
FAFARD REUBEN - Classroom: 107
COOL REUBEN - Classroom: 107
CHIONCHIO JONATHAN - Classroom: 107
CHIONCHIO BENITO - Classroom: 107
HAMER GAVIN - Classroom: 107
ALPERT GAVIN - Classroom: 107
COOL PERLA - Classroom: 107
ALPERT PERLA - Classroom: 107
HANTZ JAE - Classroom: 107
CHIONCHIO ROCIO - Classroom: 107
CHIONCHIO JED - Classroom: 107
CHIONCHIO GAVIN - Classroom: 107
FALKER ROCIO - Classroom: 107
FALKER BENITO - Classroom: 107
Search Time: 0.0156971 seconds
Enter command: C 3

```

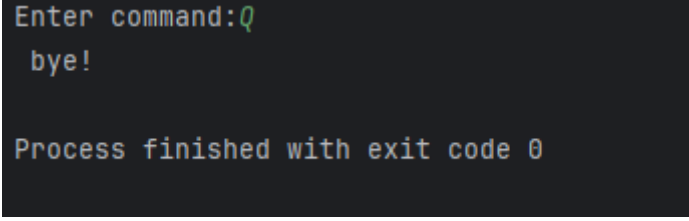
Рисунок 5 - Скріншот виконання пошуку студентів по класній кімнаті


```

Enter command:6 3
COOKUS, XUAN - Grade: 3, Classroom: 107, Bus: 52
ELHADDAD, SHANTE - Grade: 3, Classroom: 107, Bus: 52
BRODERSEN, HYE - Grade: 3, Classroom: 110, Bus: 0
SWEDLUND, SHARRI - Grade: 3, Classroom: 107, Bus: 55
CIGANEK, MANIE - Grade: 3, Classroom: 107, Bus: 53
COVINGTON, TOMAS - Grade: 3, Classroom: 107, Bus: 51
EARLY, TORY - Grade: 3, Classroom: 107, Bus: 52
LINHART, LELA - Grade: 3, Classroom: 107, Bus: 53
THEUNISSEN, GRACE - Grade: 3, Classroom: 110, Bus: 54
CREMEANS, RANDOLPH - Grade: 3, Classroom: 108, Bus: 52
COOKUS, XUAN - Grade: 3, Classroom: 104, Bus: 55
BOYTER, TOMAS - Grade: 3, Classroom: 103, Bus: 52
CREMEANS, WAN - Grade: 3, Classroom: 112, Bus: 53
CORONADO, ZANDRA - Grade: 3, Classroom: 105, Bus: 51
COVINGTON, WAN - Grade: 3, Classroom: 109, Bus: 55
COVINGTON, DIMPLE - Grade: 3, Classroom: 106, Bus: 53
CREMEANS, CARTER - Grade: 3, Classroom: 112, Bus: 56
BUSSMANN, TAMESHA - Grade: 3, Classroom: 111, Bus: 54
COVINGTON, FLOY - Grade: 3, Classroom: 105, Bus: 55
COMO, DIMPLE - Grade: 3, Classroom: 106, Bus: 56
CORKER, FLOY - Grade: 3, Classroom: 102, Bus: 54
BEX, FLOY - Grade: 3, Classroom: 110, Bus: 54
CORKER, WAN - Grade: 3, Classroom: 106, Bus: 54
BRODERSEN, JANNETTE - Grade: 3, Classroom: 105, Bus: 54
COVINGTON, TOMAS - Grade: 3, Classroom: 106, Bus: 53
CORONADO, ZANDRA - Grade: 3, Classroom: 104, Bus: 53
COMO, MANIE - Grade: 3, Classroom: 108, Bus: 52
COMO, HYE - Grade: 3, Classroom: 112, Bus: 51
CREMEANS, ZANDRA - Grade: 3, Classroom: 104, Bus: 51
BERBES, WAN - Grade: 3, Classroom: 111, Bus: 52
CLECKLER, BILLY - Grade: 3, Classroom: 107, Bus: 56
BUSSMANN, RANDOLPH - Grade: 3, Classroom: 107, Bus: 54
CREMEANS, MANIE - Grade: 3, Classroom: 104, Bus: 55
CLECKLER, WAN - Grade: 3, Classroom: 103, Bus: 52
CREMEANS, CARTER - Grade: 3, Classroom: 105, Bus: 52
BEX, TAMESHA - Grade: 3, Classroom: 107, Bus: 52
BERBES, ZANDRA - Grade: 3, Classroom: 110, Bus: 52
CIGANEK, ZANDRA - Grade: 3, Classroom: 105, Bus: 52
CORONADO, FLOY - Grade: 3, Classroom: 105, Bus: 53
CLECKLER, JANNETTE - Grade: 3, Classroom: 101, Bus: 54
CLECKLER, BILLY - Grade: 3, Classroom: 105, Bus: 53
BREVO, BILLY - Grade: 3, Classroom: 110, Bus: 52
BEX, HYE - Grade: 3, Classroom: 110, Bus: 51
CREMEANS, CARTER - Grade: 3, Classroom: 111, Bus: 53
BUSSMANN, HYE - Grade: 3, Classroom: 103, Bus: 53
BOYTER, INGER - Grade: 3, Classroom: 101, Bus: 51
CREMEANS, XUAN - Grade: 3, Classroom: 105, Bus: 56
COVINGTON, MANIE - Grade: 3, Classroom: 109, Bus: 52
BERBES, XUAN - Grade: 3, Classroom: 112, Bus: 51
BEAN, WAN - Grade: 3, Classroom: 101, Bus: 52
COVINGTON, DIMPLE - Grade: 3, Classroom: 108, Bus: 53
CLECKLER, INGER - Grade: 3, Classroom: 111, Bus: 53
BEAN, ZANDRA - Grade: 3, Classroom: 103, Bus: 54
CREMEANS, BILLY - Grade: 3, Classroom: 105, Bus: 54
COOKUS, TAMESHA - Grade: 3, Classroom: 104, Bus: 53
BEAN, BILLY - Grade: 3, Classroom: 112, Bus: 55
BRODERSEN, RANDOLPH - Grade: 3, Classroom: 109, Bus: 54
COOKUS, BILLY - Grade: 3, Classroom: 111, Bus: 51
BOYTER, RANDOLPH - Grade: 3, Classroom: 108, Bus: 52
CREMEANS, TOMAS - Grade: 3, Classroom: 109, Bus: 54
BRODERSEN, BILLY - Grade: 3, Classroom: 111, Bus: 51
COOKUS, BILLY - Grade: 3, Classroom: 111, Bus: 55
BEAN, INGER - Grade: 3, Classroom: 110, Bus: 56
CREMEANS, WAN - Grade: 3, Classroom: 110, Bus: 55
BERBES, FLOY - Grade: 3, Classroom: 112, Bus: 52
COVINGTON, WAN - Grade: 3, Classroom: 109, Bus: 53
BRODERSEN, DICK - Grade: 3, Classroom: 111, Bus: 51
BEAN, XUAN - Grade: 3, Classroom: 109, Bus: 53
BERBES, HYE - Grade: 3, Classroom: 103, Bus: 53
BERBES, BILLY - Grade: 3, Classroom: 108, Bus: 56
BREVO, BILLY - Grade: 3, Classroom: 109, Bus: 55
CORONADO, MANIE - Grade: 3, Classroom: 101, Bus: 54
Search Time: 0.0718399 seconds
Enter command:6 37

```

Рисунок 6 - Скріншот виконання пошуку студентів за оцінкою



```
Enter command:Q
bye!

Process finished with exit code 0
```

Рисунок 7 - Скріншот виконання виходу

```

Enter command:6 37
FAFARD ROCIO
BODZIONY LUZ
HANTZ GAVIN
STEIB ADOLPH
ALPERT ADOLPH
GAMBREL NANCY
COOL JONATHAN
KERBS JED
NISTENDIRK NANCY
KERBS PERLA
STEIB LUZ
BODZIONY NANCY
COOL GAVIN
FAFARD LUZ
BODZIONY JAE
GAMBREL JONATHAN
NISTENDIRK GAVIN
BODZIONY PERLA
FALKER REUBEN
NISTENDIRK BENITO
GAMBREL ROCIO
ALPERT NANCY
STEIB JED
ALPERT BENITO
FAFARD REUBEN
COOL REUBEN
CHIONCHIO JONATHAN
CHIONCHIO BENITO
HAMER GAVIN
ALPERT GAVIN
COOL PERLA
ALPERT PERLA
HANTZ JAE
CHIONCHIO ROCIO
CHIONCHIO JED
CHIONCHIO GAVIN
FALKER ROCIO
FALKER BENITO
ALPERT JONATHAN
CHIONCHIO LUZ
STEIB PERLA
KERBS JAE
HAMER JONATHAN
ALPERT JAE
GAMBREL BENITO
GAMBREL REUBEN
GAMBREL PERLA
NISTENDIRK JED
FALKER JED
STEIB REUBEN
BODZIONY JED
FAFARD BENITO
FAFARD ADOLPH
GAMBREL JAE
STEIB BENITO
NISTENDIRK ROCIO
COOL ROCIO
HANTZ REUBEN
STEIB NANCY
ALPERT LUZ
COOL LUZ
HAMER PERLA
ALPERT JED
COOL JAE
CHIONCHIO ADOLPH
NISTENDIRK REUBEN
HAMER ADOLPH
ALPERT ROCIO
HAMER REUBEN
FALKER GAVIN
NISTENDIRK PERLA
HAMER BENITO
FAFARD PERLA
KERBS BENITO
Search Time: 0.0452927 seconds

```

Рисунок 8 - Скріншот пошуку вчителів за оцінкою

```

Enter command: T ROCIO FAFARD
SAELEE, DANILO - Grade: 1, Classroom: 103, Bus: 54
GARTH, JOHN - Grade: 1, Classroom: 103, Bus: 0
BOYTER, TOMAS - Grade: 3, Classroom: 103, Bus: 52
BEAN, BILLY - Grade: 5, Classroom: 103, Bus: 53
COVINGTON, DIMPLE - Grade: 1, Classroom: 103, Bus: 54
CLECKLER, INGER - Grade: 5, Classroom: 103, Bus: 52
BREVO, CARTER - Grade: 6, Classroom: 103, Bus: 54
COVINGTON, DIMPLE - Grade: 6, Classroom: 103, Bus: 55
CLECKLER, WAN - Grade: 3, Classroom: 103, Bus: 52
BREVO, JANNETTE - Grade: 4, Classroom: 103, Bus: 56
COVINGTON, RANDOLPH - Grade: 2, Classroom: 103, Bus: 51
COOKUS, INGER - Grade: 5, Classroom: 103, Bus: 51
COVINGTON, HYE - Grade: 4, Classroom: 103, Bus: 56
BEX, BILLY - Grade: 4, Classroom: 103, Bus: 51
BUSSMANN, HYE - Grade: 3, Classroom: 103, Bus: 53
CORKER, BILLY - Grade: 6, Classroom: 103, Bus: 55
BRODERSEN, MANIE - Grade: 2, Classroom: 103, Bus: 55
BEAN, ZANDRA - Grade: 3, Classroom: 103, Bus: 54
CORKER, BILLY - Grade: 1, Classroom: 103, Bus: 56
CLECKLER, MANIE - Grade: 2, Classroom: 103, Bus: 53
COMO, FLOY - Grade: 5, Classroom: 103, Bus: 53
CREMEANS, DICK - Grade: 6, Classroom: 103, Bus: 53
BERBES, HYE - Grade: 3, Classroom: 103, Bus: 53
CIGANEK, HYE - Grade: 4, Classroom: 103, Bus: 55
COMO, FLOY - Grade: 6, Classroom: 103, Bus: 54
CIGANEK, JANNETTE - Grade: 5, Classroom: 103, Bus: 56
CORKER, BILLY - Grade: 1, Classroom: 103, Bus: 52
CREMEANS, TOMAS - Grade: 6, Classroom: 103, Bus: 53
BEX, WAN - Grade: 6, Classroom: 103, Bus: 52
BEX, HYE - Grade: 6, Classroom: 103, Bus: 56
CLECKLER, INGER - Grade: 1, Classroom: 103, Bus: 52
BEX, BILLY - Grade: 5, Classroom: 103, Bus: 51
CORKER, WAN - Grade: 4, Classroom: 103, Bus: 54
CORKER, DIMPLE - Grade: 5, Classroom: 103, Bus: 52
CORKER, CARTER - Grade: 4, Classroom: 105, Bus: 53
IMMERMAN, DEVIN - Grade: 4, Classroom: 105, Bus: 52
RAPOSE, ALONZO - Grade: 4, Classroom: 105, Bus: 51
OGAS, ERVIN - Grade: 4, Classroom: 105, Bus: 54
MASSART, ELDON - Grade: 4, Classroom: 105, Bus: 0
BEX, TAMESHA - Grade: 4, Classroom: 105, Bus: 55
CORONADO, ZANDRA - Grade: 3, Classroom: 105, Bus: 51
CORONADO, INGER - Grade: 1, Classroom: 105, Bus: 53
CORKER, FLOY - Grade: 2, Classroom: 105, Bus: 54
BEAN, TAMESHA - Grade: 6, Classroom: 105, Bus: 55
COVINGTON, FLOY - Grade: 3, Classroom: 105, Bus: 55
CIGANEK, JANNETTE - Grade: 1, Classroom: 105, Bus: 54
CLECKLER, WAN - Grade: 6, Classroom: 105, Bus: 56
BRODERSEN, JANNETTE - Grade: 3, Classroom: 105, Bus: 54
CREMEANS, CARTER - Grade: 4, Classroom: 105, Bus: 53
BEAN, BILLY - Grade: 5, Classroom: 105, Bus: 51
BRODERSEN, DIMPLE - Grade: 1, Classroom: 105, Bus: 53
COOKUS, JANNETTE - Grade: 5, Classroom: 105, Bus: 51
BEAN, WAN - Grade: 6, Classroom: 105, Bus: 52
COMO, INGER - Grade: 4, Classroom: 105, Bus: 52
BOYTER, WAN - Grade: 5, Classroom: 105, Bus: 55
CORONADO, HYE - Grade: 4, Classroom: 105, Bus: 55
BREVO, CARTER - Grade: 2, Classroom: 105, Bus: 54
COMO, INGER - Grade: 2, Classroom: 105, Bus: 53
CORONADO, DICK - Grade: 2, Classroom: 105, Bus: 55
CREMEANS, CARTER - Grade: 3, Classroom: 105, Bus: 52
BERBES, TOMAS - Grade: 6, Classroom: 105, Bus: 52
COOKUS, XUAN - Grade: 1, Classroom: 105, Bus: 52
CIGANEK, ZANDRA - Grade: 3, Classroom: 105, Bus: 52
CORONADO, FLOY - Grade: 3, Classroom: 105, Bus: 53
CLECKLER, BILLY - Grade: 3, Classroom: 105, Bus: 53
BUSSMANN, XUAN - Grade: 2, Classroom: 105, Bus: 53
BUSSMANN, TOMAS - Grade: 1, Classroom: 105, Bus: 54
CIGANEK, HYE - Grade: 5, Classroom: 105, Bus: 56
COVINGTON, JANNETTE - Grade: 4, Classroom: 105, Bus: 54
COOKUS, XUAN - Grade: 3, Classroom: 107, Bus: 52
ELHADDAD, SHANTE - Grade: 3, Classroom: 107, Bus: 52
SWEDLUND, SHARRI - Grade: 3, Classroom: 107, Bus: 55
CIGANEK, MANIE - Grade: 3, Classroom: 107, Bus: 53
COVINGTON, TOMAS - Grade: 3, Classroom: 107, Bus: 51
EARLY, TORY - Grade: 3, Classroom: 107, Bus: 52
LINHART, LELA - Grade: 3, Classroom: 107, Bus: 53
BERBES, DICK - Grade: 4, Classroom: 107, Bus: 52
CORONADO, INGER - Grade: 5, Classroom: 107, Bus: 54
CORONADO, JANNETTE - Grade: 5, Classroom: 107, Bus: 51
CORKER, TAMESHA - Grade: 2, Classroom: 107, Bus: 52
BOYTER, WAN - Grade: 1, Classroom: 107, Bus: 54
BEX, JANNETTE - Grade: 2, Classroom: 107, Bus: 51
CIGANEK, JANNETTE - Grade: 4, Classroom: 107, Bus: 54
COMO, FLOY - Grade: 4, Classroom: 107, Bus: 54
CORKER, JANNETTE - Grade: 5, Classroom: 107, Bus: 51
COVINGTON, DICK - Grade: 1, Classroom: 107, Bus: 54
CLECKLER, BILLY - Grade: 3, Classroom: 107, Bus: 56
BUSSMANN, RANDOLPH - Grade: 3, Classroom: 107, Bus: 54
COOKUS, HYE - Grade: 5, Classroom: 107, Bus: 56
BEX, TAMESHA - Grade: 3, Classroom: 107, Bus: 52
BEAN, DICK - Grade: 1, Classroom: 107, Bus: 52
BEX, HYE - Grade: 1, Classroom: 107, Bus: 55
CORKER, ZANDRA - Grade: 1, Classroom: 107, Bus: 51
BUSSMANN, HYE - Grade: 2, Classroom: 107, Bus: 54
COMO, DIMPLE - Grade: 2, Classroom: 107, Bus: 53
COVINGTON, CARTER - Grade: 4, Classroom: 107, Bus: 54
BUSSMANN, XUAN - Grade: 6, Classroom: 107, Bus: 52
BEAN, FLOY - Grade: 1, Classroom: 107, Bus: 52
BREVO, RANDOLPH - Grade: 6, Classroom: 107, Bus: 54
BOYTER, XUAN - Grade: 1, Classroom: 107, Bus: 54
BOYTER, CARTER - Grade: 2, Classroom: 107, Bus: 54
BOYTER, RANDOLPH - Grade: 1, Classroom: 107, Bus: 53
CORKER, TAMESHA - Grade: 5, Classroom: 107, Bus: 53
SCHOENECKER, PHUONG - Grade: 6, Classroom: 109, Bus: 0
CLECKLER, FLOY - Grade: 6, Classroom: 109, Bus: 53
KRAKE, VALENTIN - Grade: 6, Classroom: 109, Bus: 56
CREMEANS, RANDOLPH - Grade: 6, Classroom: 109, Bus: 0
KREESE, CARRIE - Grade: 6, Classroom: 109, Bus: 0
BREVO, FLOY - Grade: 5, Classroom: 109, Bus: 56
COVINGTON, WAN - Grade: 3, Classroom: 109, Bus: 55
BEAN, RANDOLPH - Grade: 6, Classroom: 109, Bus: 51
BERBES, TAMESHA - Grade: 6, Classroom: 109, Bus: 51
COOKUS, WAN - Grade: 5, Classroom: 109, Bus: 56
BERBES, JANNETTE - Grade: 6, Classroom: 109, Bus: 52
COOKUS, TOMAS - Grade: 4, Classroom: 109, Bus: 54
BUSSMANN, XUAN - Grade: 1, Classroom: 109, Bus: 55
COVINGTON, FLOY - Grade: 5, Classroom: 109, Bus: 56
CORONADO, BILLY - Grade: 4, Classroom: 109, Bus: 52
BUSSMANN, ZANDRA - Grade: 5, Classroom: 109, Bus: 52
CORKER, BILLY - Grade: 2, Classroom: 109, Bus: 52
COMO, CARTER - Grade: 4, Classroom: 109, Bus: 53
BEAN, JANNETTE - Grade: 2, Classroom: 109, Bus: 55
BRODERSEN, HYE - Grade: 4, Classroom: 109, Bus: 53
COVINGTON, MANIE - Grade: 3, Classroom: 109, Bus: 52
CLECKLER, JANNETTE - Grade: 4, Classroom: 109, Bus: 53
CIGANEK, FLOY - Grade: 2, Classroom: 109, Bus: 53
COMO, FLOY - Grade: 1, Classroom: 109, Bus: 53
CLECKLER, MANIE - Grade: 5, Classroom: 109, Bus: 51
BRODERSEN, RANDOLPH - Grade: 3, Classroom: 109, Bus: 54
CLECKLER, HYE - Grade: 4, Classroom: 109, Bus: 55
CREMEANS, TOMAS - Grade: 3, Classroom: 109, Bus: 54
COVINGTON, FLOY - Grade: 5, Classroom: 109, Bus: 51
CORKER, ZANDRA - Grade: 2, Classroom: 109, Bus: 53
COVINGTON, WAN - Grade: 3, Classroom: 109, Bus: 53
CIGANEK, DIMPLE - Grade: 5, Classroom: 109, Bus: 54
BEX, WAN - Grade: 2, Classroom: 109, Bus: 53
COOKUS, BILLY - Grade: 6, Classroom: 109, Bus: 51
BEAN, XUAN - Grade: 3, Classroom: 109, Bus: 53
COMO, WAN - Grade: 4, Classroom: 109, Bus: 54
BREVO, BILLY - Grade: 3, Classroom: 109, Bus: 55
BEAN, TAMESHA - Grade: 6, Classroom: 109, Bus: 54
CORKER, JANNETTE - Grade: 5, Classroom: 109, Bus: 56
Search Time: 0.0906228 seconds

```

Рисунок 9 - Скріншот пошуку студентів за вчителем

```

Enter command:S K
K, DAN - Grade: 12, Classroom: 108, Bus: 0, Teachers: HAMER GAVIN; KERBS ADOLPH; COOL ADOLPH; H
ANTZ NANCY; FALKER GAVIN; HANTZ BENITO; STEIB BENITO; CHIONCHIO JONATHAN; FALKER ROCIO; HANTZ
; FALKER JAE; HANTZ GAVIN; FALKER ADOLPH; NISTENDIRK JED; CHIONCHIO BENITO; FALKER NANCY; COOL
TO; CHIONCHIO JAE; GAMBREL JONATHAN; STEIB REUBEN; NISTENDIRK JONATHAN; BODZIONY LUZ; HAMER BENI
DZIONY JAE; CHIONCHIO ADOLPH; ALPERT NANCY; ALPERT REUBEN; ALPERT ADOLPH; STEIB JONATHAN; GAMBRE
ANCY; KERBS JONATHAN; COOL PERLA; HAMER JONATHAN; NISTENDIRK ADOLPH;
Enter command:DEL S
Enter last name:K
Enter first name:DAN
Student deleted.
Enter command:S K
No students found with last name: K

```

Рисунок 10 - Видалення студента

```

Enter command:C 107T
FAFARD ROCIO - Classroom: 107
BODZIONY LUZ - Classroom: 107
HANTZ GAVIN - Classroom: 107
STEIB ADOLPH - Classroom: 107
ALPERT ADOLPH - Classroom: 107
GAMBREL NANCY - Classroom: 107
COOL JONATHAN - Classroom: 107
KERBS JED - Classroom: 107
NISTENDIRK NANCY - Classroom: 107
KERBS PERLA - Classroom: 107
STEIB LUZ - Classroom: 107
BODZIONY NANCY - Classroom: 107
COOL GAVIN - Classroom: 107
FAFARD LUZ - Classroom: 107
BODZIONY JAE - Classroom: 107
GAMBREL JONATHAN - Classroom: 107
NISTENDIRK GAVIN - Classroom: 107
BODZIONY PERLA - Classroom: 107
FALKER REUBEN - Classroom: 107
NISTENDIRK BENITO - Classroom: 107
GAMBREL ROCIO - Classroom: 107
ALPERT NANCY - Classroom: 107
STEIB JED - Classroom: 107
ALPERT BENITO - Classroom: 107
FAFARD REUBEN - Classroom: 107
COOL REUBEN - Classroom: 107
CHIONCHIO JONATHAN - Classroom: 107
CHIONCHIO BENITO - Classroom: 107
HAMER GAVIN - Classroom: 107
ALPERT GAVIN - Classroom: 107
COOL PERLA - Classroom: 107
ALPERT PERLA - Classroom: 107
HANTZ JAE - Classroom: 107
CHIONCHIO ROCIO - Classroom: 107
CHIONCHIO JED - Classroom: 107
CHIONCHIO GAVIN - Classroom: 107
FALKER ROCIO - Classroom: 107
FALKER BENITO - Classroom: 107
KARAS IRYNA - Classroom: 107
Enter command:DEL T
Enter last name:KARAS
Enter first name:IRYNA
Teacher deleted.

```

Рисунок 11 - Видалення викладача

```

Enter command:ADD S
Enter last name:KIRIIENKO
Enter first name:DANYLO
Enter grade:12
Enter classroom:120
Enter bus number:1
Enter command:ADD T
Enter l
ast name:KARAS
Enter first name:IRYNA
Enter classroom:120
Enter command:S KIRIIENKO
KIRIIENKO, DANYLO - Grade: 12, Classroom: 120, Bus: 1, Teach
ers: KARAS IRYNA;
Enter command:EDIT S
Enter last name:KIRIIENKO
Editing student KIRIIENKO, DANYLO
Enter new grade:25
Enter new classroom:125
Enter new bus number:02
Enter command:EDIT T
Enter last name:KARAS
Editing teacher KARAS, IRYNA

Enter new classroom:125
Teacher and students updated.
Enter command: S KIRIIENKO
Error! Invalid command!
Enter command:S KIRIIENKO
KIRIIENKO, DANYLO - Grade: 25, Classroom: 125, Bus: 2, Teachers: KARAS IRYNA;
Enter command:

```

Рисунок 12 - Додавання і редакція студента та викладача

Висновки:

Я написав програму, яка здійснює пошук студентів у текстовому файлі та видає результати пошуку. Також студентів і викладачів можна видаляти, змінювати і додавати нових (всі дії записуються в файлі) Її можна переглянути за посиланням нижче:

<https://github.com/ddanq4/3-obd/blob/main/2/main.cpp>

Текст програми з коментарями:*main.cpp*

```

1 #include <iostream>
2 #include <string>
3 #include <vector>
4 #include <fstream>
5 #include <sstream>
6 #include <map>
7 #include <algorithm>
8 #include <set>
9
10 std::string trim(const std::string& str) {
11     size_t first = str.find_first_not_of(' ');
12     size_t last = str.find_last_not_of(' ');
13     return (first == std::string::npos || last == std::string::npos) ? "" :
14     str.substr(first, last - first + 1);
15 }
16
17 struct Student {
18     std::string lastName;
19     std::string firstName;
20     int grade;
21     int classroom;
22     int bus;
23 };
24
25 struct Teacher {
26     std::string lastName;
27     std::string firstName;
28     int classroom;
29 };
30
31 std::vector<Student> loadStudents(const std::string& filename) {
32     std::vector<Student> students;
33     std::ifstream file(filename);
34     std::string line;
35
36     if (!file.is_open()) {
37         std::cerr << "Не удалось открыть файл: " << filename << std::endl;
38         return students;
39     }
40
41     while (std::getline(file, line)) {
42         std::stringstream ss(line);
43         Student student;
44         std::getline(ss, student.lastName, ',');
45         std::getline(ss, student.firstName, ',');
46         ss >> student.grade;
47         ss.ignore();

```



```

48     ss >> student.classroom;
49     ss.ignore();
50     ss >> student.bus;
51     students.push_back(student);
52 }
53
54     return students;
55 }
56
57 std::multimap<int, Teacher> loadTeachers(const std::string& filename) {
58     std::multimap<int, Teacher> teachers;
59     std::ifstream file(filename);
60     std::string line;
61
62     if (!file.is_open()) {
63         std::cerr << "Не удалось открыть файл: " << filename << std::endl;
64         return teachers;
65     }
66
67     while (std::getline(file, line)) {
68         std::stringstream ss(line);
69         Teacher teacher;
70         std::getline(ss, teacher.lastName, ',');
71         std::getline(ss, teacher.firstName, ',');
72         int classroom;
73         ss >> classroom;
74         teachers.insert({classroom, teacher});
75     }
76
77     return teachers;
78 }
79
80 void saveStudents(const std::vector<Student>& students, const std::string&
81 filename) {
82     std::ofstream file(filename);
83     if (!file.is_open()) {
84         std::cerr << "Не удалось открыть файл для записи: " << filename <<
85 std::endl;
86         return;
87     }
88     for (const auto& student : students) {
89         file << student.lastName << ", " << student.firstName << ", "
90             << student.grade << ", " << student.classroom << ", " <<
91 student.bus << "\n";
92     }
93 }
94
95 void saveTeachers(const std::multimap<int, Teacher>& teachers, const
96 std::string& filename) {
97     std::ofstream file(filename);

```

```

98     if (!file.is_open()) {
99         std::cerr << "Не удалось открыть файл для записи: " << filename <<
100 std::endl;
101         return;
102     }
103     for (const auto& teacher : teachers) {
104         file << teacher.second.lastName << ", " << teacher.second.firstName
105 << ", " << teacher.first << "\n";
106     }
107 }
108
109 void addStudent(std::vector<Student>& students) {
110     Student newStudent;
111     std::cout << "Enter last name: ";
112     std::cin >> newStudent.lastName;
113     newStudent.lastName = trim(newStudent.lastName);
114     std::cout << "Enter first name: ";
115     std::cin >> newStudent.firstName;
116     newStudent.firstName = trim(newStudent.firstName);
117     std::cout << "Enter grade: ";
118     std::cin >> newStudent.grade;
119     std::cout << "Enter classroom: ";
120     std::cin >> newStudent.classroom;
121     std::cout << "Enter bus number: ";
122     std::cin >> newStudent.bus;
123     students.push_back(newStudent);
124     std::cin.ignore();
125 }
126
127 void deleteStudent(std::vector<Student>& students, const std::string&
128 lastName, const std::string& firstName) {
129     auto it = std::remove_if(students.begin(), students.end(), [&lastName,
130 &firstName](const Student& student) {
131         return trim(student.lastName) == trim(lastName) &&
132 trim(student.firstName) == trim(firstName);
133     });
134     if (it != students.end()) {
135         students.erase(it, students.end());
136         std::cout << "Student deleted.\n";
137     } else {
138         std::cout << "Student not found.\n";
139     }
140 }
141
142 void editStudent(std::vector<Student>& students, const std::string&
143 lastName) {
144     auto it = std::find_if(students.begin(), students.end(),
145 [&lastName](const Student& student) {
146         return trim(student.lastName) == trim(lastName);
147     });

```

```

148     if (it != students.end()) {
149         std::cout << "Editing student " << it->lastName << ", " << it-
150 >firstName << "\n";
151         std::cout << "Enter new grade: ";
152         std::cin >> it->grade;
153         std::cout << "Enter new classroom: ";
154         std::cin >> it->classroom;
155         std::cout << "Enter new bus number: ";
156         std::cin >> it->bus;
157     } else {
158         std::cout << "Student not found.\n";
159     }
160 }
161
162 void addTeacher(std::multimap<int, Teacher>& teachers) {
163     Teacher newTeacher;
164     std::cout << "Enter last name: ";
165     std::cin >> newTeacher.lastName;
166     newTeacher.lastName = trim(newTeacher.lastName);
167     std::cout << "Enter first name: ";
168     std::cin >> newTeacher.firstName;
169     newTeacher.firstName = trim(newTeacher.firstName);
170     std::cout << "Enter classroom: ";
171     std::cin >> newTeacher.classroom;
172     teachers.insert({newTeacher.classroom, newTeacher});
173 }
174
175 void deleteTeacher(std::multimap<int, Teacher>& teachers, const
176 std::string& lastName, const std::string& firstName) {
177     auto it = std::find_if(teachers.begin(), teachers.end(), [&lastName,
178 &firstName](const std::pair<int, Teacher>& teacher) {
179         return trim(teacher.second.lastName) == trim(lastName) &&
180 trim(teacher.second.firstName) == trim(firstName);
181     });
182     if (it != teachers.end()) {
183         teachers.erase(it);
184         std::cout << "Teacher deleted.\n";
185     } else {
186         std::cout << "Teacher not found.\n";
187     }
188 }
189
190 void editTeacher(std::multimap<int, Teacher>& teachers, const std::string&
191 lastName, std::vector<Student>& students, const std::string& teachersFile)
192 {
193     auto it = std::find_if(teachers.begin(), teachers.end(),
194 [&lastName](const std::pair<int, Teacher>& teacher) {
195         return trim(teacher.second.lastName) == trim(lastName);
196     });
197

```

```

198     if (it != teachers.end()) {
199         int oldClassroom = it->first;
200         Teacher updatedTeacher = it->second;
201
202         std::cout << "Editing teacher " << it->second.lastName << ", " <<
203 it->second.firstName << "\n";
204         std::cout << "Enter new classroom: ";
205         std::cin >> updatedTeacher.classroom;
206         teachers.erase(it);
207         teachers.insert({updatedTeacher.classroom, updatedTeacher});
208         for (auto& student : students) {
209             if (student.classroom == oldClassroom) {
210                 student.classroom = updatedTeacher.classroom;
211             }
212         }
213         saveTeachers(teachers, teachersFile);
214         std::cout << "Teacher and students updated.\n";
215     } else {
216         std::cout << "Teacher not found.\n";
217     }
218 }
219
220
221 void searchByStudent(const std::vector<Student>& students, const
222 std::string& lastName, const std::multimap<int, Teacher>& teachers) {
223     bool found = false;
224     for (const auto& student : students) {
225         if (trim(student.lastName) == trim(lastName)) {
226             auto range = teachers.equal_range(student.classroom);
227             std::cout << student.lastName << ", " << student.firstName
228                 << " - Grade: " << student.grade
229                 << ", Classroom: " << student.classroom
230                 << ", Bus: " << student.bus
231                 << ", Teachers: ";
232             for (auto it = range.first; it != range.second; ++it) {
233                 std::cout << it->second.lastName << " " << it-
234 >second.firstName << "; ";
235             }
236             std::cout << std::endl;
237             found = true;
238         }
239     }
240
241     if (!found) {
242         std::cout << "No students found with last name: " << lastName <<
243 std::endl;
244     }
245 }
246
247 void searchByBus(const std::vector<Student>& students, int busNumber) {

```

```

248     bool found = false;
249     for (const auto& student : students) {
250         if (student.bus == busNumber) {
251             std::cout << student.lastName << ", " << student.firstName
252                 << " - Bus: " << student.bus
253                 << std::endl;
254             found = true;
255         }
256     }
257     if (!found) {
258         std::cout << "No students found on bus number: " << busNumber <<
259 std::endl;
260     }
261 }
262
263 void searchByClassroom(const std::vector<Student>& students, int classroom)
264 {
265     bool found = false;
266     for (const auto& student : students) {
267         if (student.classroom == classroom) {
268             std::cout << student.lastName << ", " << student.firstName
269                 << " - Classroom: " << student.classroom
270                 << std::endl;
271             found = true;
272         }
273     }
274
275     if (!found) {
276         std::cout << "No students found in classroom: " << classroom <<
277 std::endl;
278     }
279 }
280
281 void searchTeacherByClassroom(const std::multimap<int, Teacher>& teachers,
282 int classroom) {
283     auto range = teachers.equal_range(classroom);
284     if (range.first != range.second) {
285         for (auto it = range.first; it != range.second; ++it) {
286             std::cout << it->second.lastName << " " << it->second.firstName
287                 << " - Classroom: " << classroom << std::endl;
288         }
289     } else {
290         std::cout << "No teachers found for classroom: " << classroom <<
291 std::endl;
292     }
293 }
294
295 void searchTeachersByGrade(const std::vector<Student>& students, const
296 std::multimap<int, Teacher>& teachers, int grade) {
297     std::set<std::string> displayedTeachers;

```

```

298     for (const auto& student : students) {
299         if (student.grade == grade) {
300             auto range = teachers.equal_range(student.classroom);
301             for (auto it = range.first; it != range.second; ++it) {
302                 std::string fullName = it->second.lastName + " " + it->
303 second.firstName;
304                 if (displayedTeachers.find(fullName) ==
305 displayedTeachers.end()) {
306                     std::cout << it->second.lastName << " " << it->
307 second.firstName << std::endl;
308                     displayedTeachers.insert(fullName);
309                 }
310             }
311         }
312     }
313
314     if (displayedTeachers.empty()) {
315         std::cout << "No teachers found for grade: " << grade << std::endl;
316     }
317 }
318
319 void searchByGrade(const std::vector<Student>& students, int grade) {
320     bool found = false;
321     for (const auto& student : students) {
322         if (student.grade == grade) {
323             std::cout << student.lastName << ", " << student.firstName
324                 << " - Grade: " << student.grade
325                 << ", Classroom: " << student.classroom
326                 << ", Bus: " << student.bus
327                 << std::endl;
328             found = true;
329         }
330     }
331
332     if (!found) {
333         std::cout << "No students found for grade: " << grade << std::endl;
334     }
335 }
336
337 void searchStudentsByTeacher(const std::vector<Student>& students, const
338 std::multimap<int, Teacher>& teachers, const std::string& fullName) {
339     bool found = false;
340
341     for (const auto& teacher : teachers) {
342         std::string teacherFullName = teacher.second.lastName + " " +
343 teacher.second.firstName;
344
345         if (teacherFullName == fullName) {
346             for (const auto& student : students) {
347                 if (student.classroom == teacher.first) {

```

```

348         std::cout << student.lastName << ", " <<
349 student.firstName
350         << " - Grade: " << student.grade
351         << ", Classroom: " << student.classroom
352         << ", Bus: " << student.bus
353         << std::endl;
354         found = true;
355     }
356 }
357 }
358 }
359
360 if (!found) {
361     std::cout << "No students found for teacher: " << fullName <<
362 std::endl;
363 }
364 }
365
366 void commandLoop(std::vector<Student>& students, std::multimap<int,
367 Teacher>& teachers, const std::string& studentsFile, const std::string&
368 teachersFile) {
369     std::string command;
370     std::cout << "Enter command: ";
371
372     while (std::getline(std::cin, command)) {
373         if (command == "Q" || command == "Quit") break;
374
375         if (command[0] == 'S') {
376             std::string lastName = command.substr(2);
377             searchByStudent(students, lastName, teachers);
378         } else if (command[0] == 'B') {
379             int busNumber = std::stoi(command.substr(2));
380             searchByBus(students, busNumber);
381         } else if (command[0] == 'C' && command.back() == 'T') {
382             int classroom = std::stoi(command.substr(2, command.size() -
383 3));
384             searchTeacherByClassroom(teachers, classroom);
385         } else if (command[0] == 'C') {
386             int classroom = std::stoi(command.substr(2));
387             searchByClassroom(students, classroom);
388         } else if (command[0] == 'G' && command.back() == 'T') {
389             int grade = std::stoi(command.substr(2, command.size() - 3));
390             searchTeachersByGrade(students, teachers, grade);
391         } else if (command[0] == 'G') {
392             int grade = std::stoi(command.substr(2));
393             searchByGrade(students, grade);
394         } else if (command[0] == 'T') {
395             std::string fullName = command.substr(2);
396             searchStudentsByTeacher(students, teachers, fullName);
397         } else if (command == "ADD S") {

```



```

398         addStudent(students);
399         saveStudents(students, studentsFile);
400     } else if (command == "DEL S") {
401         std::string lastName, firstName;
402         std::cout << "Enter last name: ";
403         std::cin >> lastName;
404         std::cout << "Enter first name: ";
405         std::cin >> firstName;
406         deleteStudent(students, lastName, firstName);
407         saveStudents(students, studentsFile);
408         std::cin.ignore();
409     } else if (command == "EDIT S") {
410         std::string lastName;
411         std::cout << "Enter last name: ";
412         std::cin >> lastName;
413         editStudent(students, lastName);
414         saveStudents(students, studentsFile);
415         std::cin.ignore();
416     } else if (command == "ADD T") {
417         addTeacher(teachers);
418         saveTeachers(teachers, teachersFile);
419         std::cin.ignore();
420     } else if (command == "DEL T") {
421         std::string lastName, firstName;
422         std::cout << "Enter last name: ";
423         std::cin >> lastName;
424         std::cout << "Enter first name: ";
425         std::cin >> firstName;
426         deleteTeacher(teachers, lastName, firstName);
427         saveTeachers(teachers, teachersFile);
428         std::cin.ignore();
429     } else if (command == "EDIT T") {
430         std::string lastName;
431         std::cout << "Enter last name: ";
432         std::cin >> lastName;
433         editTeacher(teachers, lastName, students, teachersFile);
434         saveStudents(students, studentsFile);
435         std::cin.ignore();
436     } else {
437         std::cout << "Error! Invalid command!" << std::endl;
438     }
439     std::cout << "Enter command: ";
440 }

int main() {
    auto students = loadStudents("list.txt");
    auto teachers = loadTeachers("teachers.txt");

    if (students.empty() || teachers.empty()) {

```

```
        std::cout << "Error loading data." << std::endl;
        return 1;
    }

    commandLoop(students, teachers, "list.txt", "teachers.txt");
    std::cout << "bye!" << std::endl;
    return 0;
}
```